Exploratory Data Analytics

(E1CSA231)

Micro Project Report

On

SUBSCRIPTION CHURN ANALYSIS

Submitted by

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Abstract

This project analyzes the Telco Customer Churn Dataset obtained from Kaggle to understand why customers leave subscription-based services. The analysis was carried out using Python along with libraries such as Pandas, NumPy, Matplotlib, and Seaborn in Google Colab for data cleaning, processing, and visualization. The primary objective was to examine customer data and identify key factors influencing churn, including contract type, tenure, monthly charges, and payment methods. By exploring these variables, the study aims to uncover patterns in customer behavior and provide insights that can support effective customer retention strategies.

The findings reveal that approximately 26% of customers have churned, indicating a significant loss. Customers on month-to-month contracts were more likely to leave compared to those with long-term agreements. Additionally, customers with shorter tenure showed higher churn rates, suggesting that newer customers are more prone to discontinuing the service. Higher monthly charges were also associated with an increased likelihood of churn, and customers using electronic check payment methods exhibited comparatively higher churn rates. Overall, the analysis highlights critical factors affecting customer retention and provides valuable insights for improving customer satisfaction and reducing churn.

Overall, this analysis emphasizes the importance of proactive customer retention strategies in subscription-based businesses. By identifying high-risk customer segments—such as those with short tenure, higher monthly charges, and flexible contract types—organizations can implement targeted interventions like personalized offers, improved onboarding experiences, and incentives for long-term commitments. Additionally, optimizing payment experiences and addressing potential service dissatisfaction can further reduce churn. These insights enable businesses to make data-driven decisions, enhance customer satisfaction, and ultimately improve long-term profitability.

1. Introduction

1.1 Project Overview

The dataset used for this analysis is the **Telco Customer Churn Dataset**, which is publicly available on Kaggle. It contains detailed information about customers, including demographic data (such as gender, senior citizen status, and dependents), services subscribed (like internet service, phone service, and additional features), and account-related details (such as tenure, contract type, monthly charges, and payment methods). The dataset also includes a target variable indicating whether a customer has churned or not, making it suitable for both exploratory data analysis and predictive modeling.

The analysis begins with data cleaning and preprocessing, where missing values are handled and categorical variables are properly encoded. After preparing the data, exploratory data analysis is performed using statistical methods and visualizations to uncover patterns and relationships between different variables and customer churn. Tools such as bar charts, histograms, and heatmaps are used to clearly present these patterns.

Through this analysis, several important insights are identified. It is observed that customers with month-to-month contracts, shorter tenure, and higher monthly charges are more likely to churn. Additionally, certain payment methods are associated with higher churn rates. Overall, the project provides a comprehensive understanding of customer churn behavior and offers valuable insights that can assist organizations in developing effective strategies to reduce churn and improve customer loyalty.

This project is about analyzing customer churn in a telecom company to understand why customers stop using the service. The dataset used is the Telco Customer Churn Dataset from Kaggle, which includes details like customer demographics, services, contract types, tenure, and payment methods. The analysis involves cleaning the data and using simple visualizations to find patterns related to churn. It helps identify which types of customers are more likely to leave. Overall, the project gives a clear understanding of customer behavior and supports better decision-making to improve customer retention.

1.2 Objectives of the Project

- To analyze customer data in order to gain a clear understanding of churn trends and overall customer behavior
- To measure the proportion of customers who have discontinued the service and evaluate its business impact
- To identify and assess the key factors influencing customer churn, including contract type, tenure, and monthly charges
- To examine the relationship between customer demographics and their likelihood of churn
- To evaluate how different services and subscription features contribute to customer retention or attrition
- To analyze the influence of pricing and payment methods on customer decisions
- To compare churn patterns across different customer segments for deeper insights
- To uncover meaningful relationships between variables using data analysis and visualization techniques
- To generate actionable insights that support data-driven decision-making
- To recommend strategies aimed at reducing churn and improving long-term customer retention

1.3 Problem Statement

Customer churn is a major challenge for telecom companies as it directly affects revenue, profitability, and customer base stability. The problem addressed in this project is to identify and analyze the key factors that lead customers to discontinue their services. High churn not only reflects customer dissatisfaction but also increases the cost of acquiring new customers. This project aims to study customer data to uncover patterns and relationships between variables such as contract type, tenure, pricing, service usage, and demographics. By understanding these factors, the analysis helps in predicting churn behavior and supports the development of effective strategies to improve customer retention and overall service quality.

Key Factors Affecting Customer Churn

- Contract type (higher churn in month-to-month plans)
- Customer tenure (new customers are more likely to churn)
- Monthly charges (higher costs increase churn probability)
- Payment methods (electronic check shows higher churn)
- Service usage (internet and additional services influence retention)
- Customer demographics (age, dependents, etc.)
- Perceived value and customer experience

1.4 Scope of the Project

This project focuses on analyzing customer churn in a telecom company using the Telco Customer Churn dataset. The primary scope of the project is to perform exploratory data analysis (EDA) in order to understand customer behavior and identify the key factors that contribute to churn. It includes data cleaning, preprocessing, and visualization of important variables such as contract type, tenure, monthly charges, payment methods, and services subscribed by customers. The analysis aims to discover patterns, trends, and relationships between these variables and customer churn. The project also provides meaningful insights and basic recommendations that can help improve customer retention and support better business decision-making.

The scope of this project is limited to descriptive and analytical study only. It does not include the development or implementation of machine learning models for predicting customer churn. Advanced techniques such as classification algorithms, model training, testing, and evaluation are not part of this project. Additionally, the project does not involve real-time data processing, system deployment, or integration with business applications. The analysis is entirely based on a static dataset obtained from Kaggle, and no external or live data sources are used. Furthermore, the findings are limited to the dataset provided and may not fully represent all real-world scenarios. Overall, the project is confined to understanding and explaining churn behavior through data analysis rather than building predictive or automated solutions.

Included in the project:

- Exploratory Data Analysis (EDA) of the Telco Customer Churn dataset
- Data cleaning and preprocessing of customer data
- Analysis of key factors such as contract type, tenure, monthly charges, and payment methods
- Visualization of data using charts and graphs for better understanding
- Identification of patterns and relationships affecting customer churn
- Deriving insights and providing basic recommendations for customer retention

Not included in the project:

- Machine learning or predictive modeling for churn prediction
- Model training, testing, and performance evaluation
- Real-time data analysis or system deployment
- Use of external or live datasets beyond the given dataset
- Development of automated decision-making systems
- Implementation of business strategies in real-world environments

2. Dataset Description

2.1 Source of Data

The dataset used for this project is the **"Telco Customer Churn Dataset",** which has been obtained from **Kaggle**. It is a publicly available dataset commonly used for data analysis and research purposes. The data is provided in CSV (Comma-Separated Values) format, making it suitable for analysis using tools such as Python and Excel.

2.2 Dataset Overview

The Telco Customer Churn dataset consists of **7043 rows and 21 columns**, where each row represents an individual customer and each column represents a specific attribute related to that customer. The dataset provides detailed information about customer demographics, account details, services subscribed, and whether the customer has churned or not.

The dataset contains a combination of **categorical and numerical variables**. Categorical data includes attributes such as gender, senior citizen status, partner, dependents, contract type, internet service, and payment method. Numerical data includes variables like tenure (number of months a customer has stayed), monthly charges, and total charges. The target variable in the dataset is **"Churn"**, which indicates whether a customer has left the service.

Additionally, the dataset is structured and well-organized, making it suitable for data analysis and visualization. Some preprocessing steps, such as handling missing values and converting data types, are required before analysis. Overall, the dataset provides a comprehensive view of customer behavior, enabling effective analysis of factors influencing customer churn.

2.3 Features

Column Name	Description
customerID	Unique ID assigned to each customer
gender	Gender of the customer (Male/Female)
SeniorCitizen	Whether the customer is a senior citizen (0 = No, 1 = Yes)
Partner	Whether the customer has a partner
Dependents	Whether the customer has dependents
tenure	Number of months the customer has stayed with the company
PhoneService	Whether the customer has phone service
Multiple Lines	Whether the customer has multiple phone lines
Internet Service	Type of internet service (DSL, Fiber optic, None)
Online Security	Whether the customer has online security service
Online Backup	Whether the customer has online backup service
Device Protection	Whether the customer has device protection
Tech Support	Whether the customer has technical support
Streaming TV	Whether the customer uses streaming TV service
Streaming Movies	Whether the customer uses streaming movies service
Contract	Type of contract (Month-to-month, One year, Two year)
Paperless Billing	Whether the customer uses paperless billing
Payment Method	Payment method used by the customer
MonthlyCharges	Amount charged to the customer monthly

TotalCharges	Total amount charged to the customer
Churn	Whether the customer has left the service (Yes/No)

Table 2.1 Dataset Description

3. Data Preprocessing

3.1 Data Cleaning

Data cleaning was performed to ensure the dataset is accurate and suitable for analysis. Duplicate records, if any, were checked and removed to avoid redundancy. In addition, inconsistencies in data formats were corrected, particularly in the Total Charges column, which contained non-numeric values. These values were converted into a proper numeric format to enable further analysis. Overall, the dataset was cleaned to maintain consistency, accuracy, and reliability.

3.2 Handling Missing Values

The dataset contained missing or invalid values primarily in the TotalCharges column, which appeared as blank or non-numeric entries. These values were identified during the data type conversion process. To handle this issue, the invalid entries were converted into null values using appropriate techniques. Instead of filling these values with mean or median, the rows containing missing values were removed to maintain data integrity. This ensured that the analysis was performed on complete and reliable data.

3.3 Data Transformation

Data transformation was carried out to make the dataset suitable for analysis. The TotalCharges column was converted from object type to numeric format. Additionally, categorical values such as Senior Citizen were transformed into more meaningful labels (e.g., “Senior Citizen” and “Not Senior Citizen”) for better interpretation. Although full encoding techniques like one-hot encoding were not applied, categorical variables were effectively used in visualizations. No scaling or normalization techniques were applied, as the analysis focused on visualization and pattern identification rather than machine learning.

3.4 Outlier Detection

Outliers in the dataset were identified using **boxplots**, which visually highlight extreme values in numerical features such as **Monthly Charges** and **tenure**. The boxplot method helps detect values

that lie outside the typical range (based on the interquartile range – IQR). These extreme values indicate potential outliers in the dataset.

After careful analysis, no outliers were removed, as they represent valid customer data and may provide important insights into customer behavior. Retaining these values ensures that the analysis reflects real-world scenarios accurately.

4. Exploratory Data Analysis

4.1 Univariate Analysis

- The churn distribution shows that **most customers have not churned**, but a significant portion (~26%) have left, indicating a serious retention issue.
- The contract distribution shows that **month-to-month contracts dominate**, meaning most customers prefer short-term commitments.
- Monthly charges are widely spread, but a large number of customers fall in the **higher price range (70–100)**.
- Tenure distribution shows many customers are either **new (low tenure)** or **long-term (high tenure)**, indicating a gap in mid-term retention.

4.2 Bivariate Analysis

- Customers with **month-to-month contracts have the highest churn**, while one-year and two-year contracts show very low churn → strong indicator.
- **Gender has no significant impact** on churn (male and female behavior is almost identical).
- **Senior citizens have a higher churn rate proportionally**, even though their total count is lower.
- Customers using **fiber optic internet show higher churn**, possibly due to higher cost or expectations.
- Customers using **electronic check payment method have the highest churn**, which is a very strong behavioral signal.

4.3 Multivariate Analysis

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- **Gender has no significant impact** on churn (male and female behavior is almost identical).
- **Senior citizens have a higher churn rate proportionally**, even though their total count is lower.
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4.4 Data Visualization

- From pair plot and relationships:
  - TotalCharges increases with tenure → expected (longer stay = more payment).
  - Customers with low tenure and high monthly charges are more likely to churn.
  - This shows churn is not caused by one factor, but a combination of tenure + pricing.

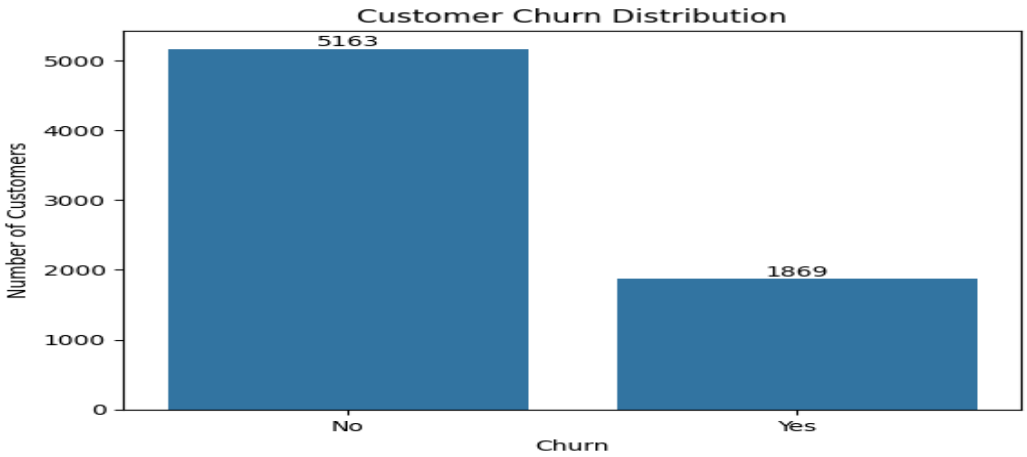
4.5 Correlation Analysis

- There is a **strong positive correlation** between:
  - tenure and TotalCharges
- There is a **weak correlation** between:
  - MonthlyCharges and tenure
- This indicates:
  - Spending increases with time
  - But **monthly price alone does not depend on tenure**

5. Results and Insights

1. Customer Churn Distribution

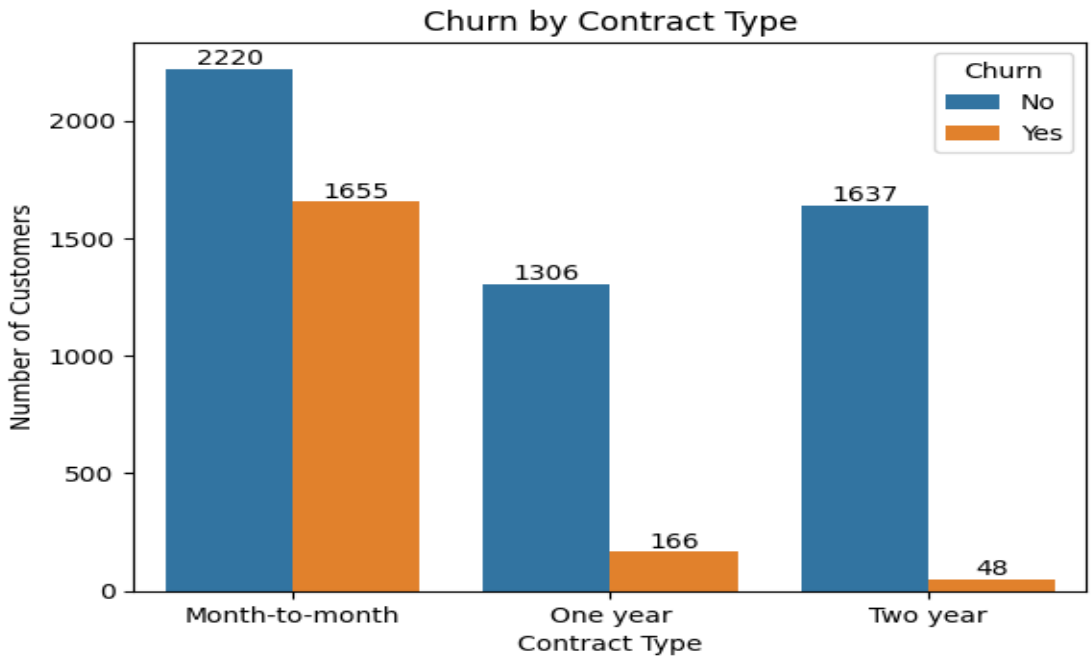
- The graph shows that a majority of customers have not churned, but a significant portion has left.
- This indicates a moderate churn rate (~26%), which is a concern for the business.



GRAPH 1

2. Churn by Contract Type

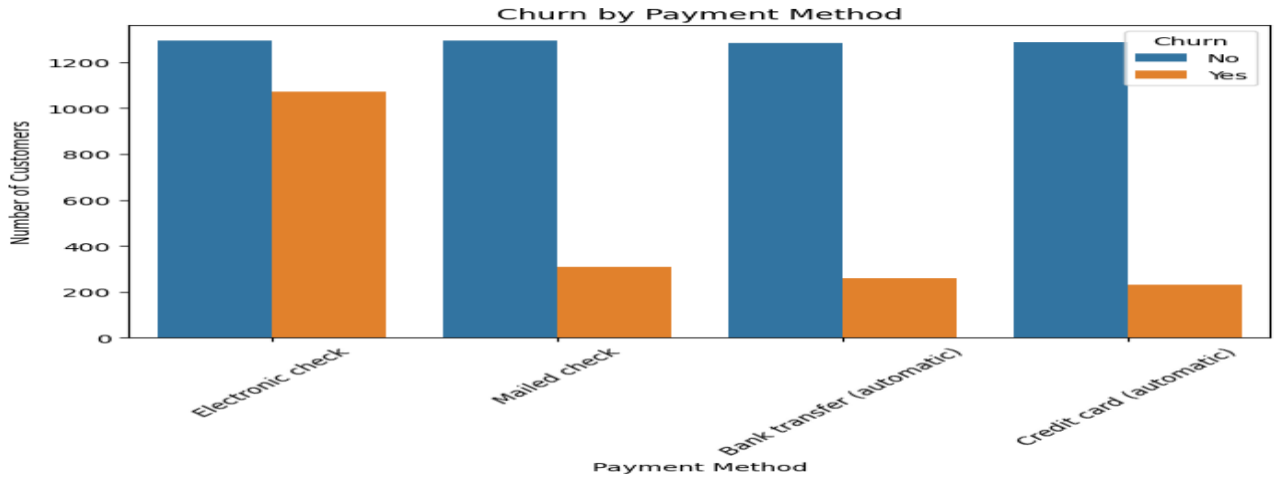
- Customers with month-to-month contracts show the highest churn.
- Long-term contracts (one-year, two-year) have significantly lower churn rates.



GRAPH 2

3. Churn by Payment Method

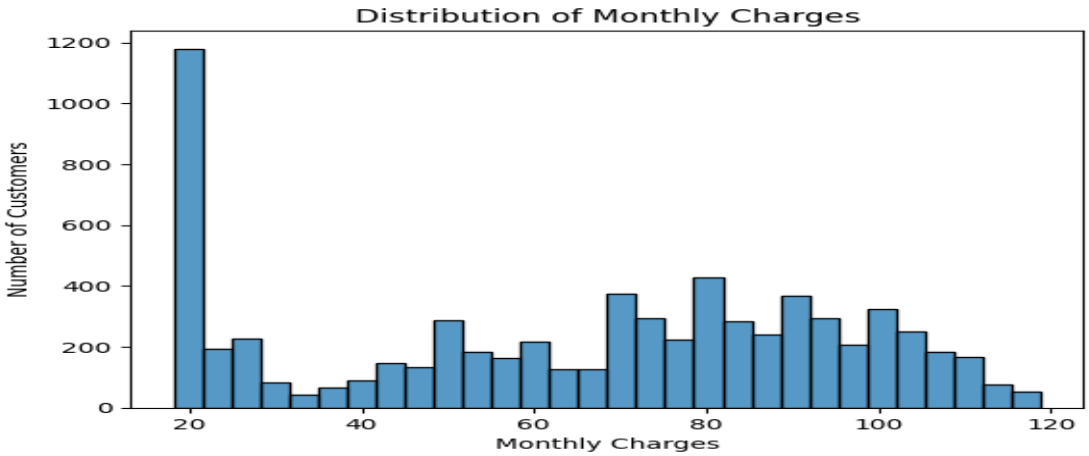
- Customers using electronic check have the highest churn rate.
- Customers using automatic payment methods show better retention.



GRAPH 3

4. Monthly Charges Distribution

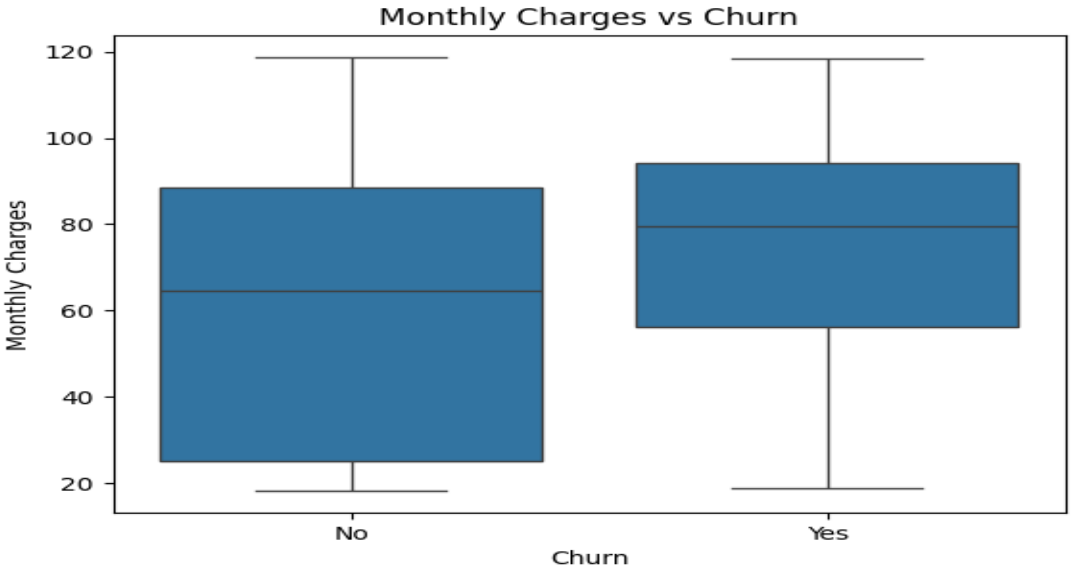
- The histogram shows that many customers fall in the higher price range.
- Higher charges are linked with higher churn.



GRAPH 4

5. Monthly Charges vs Churn (Box Plot)

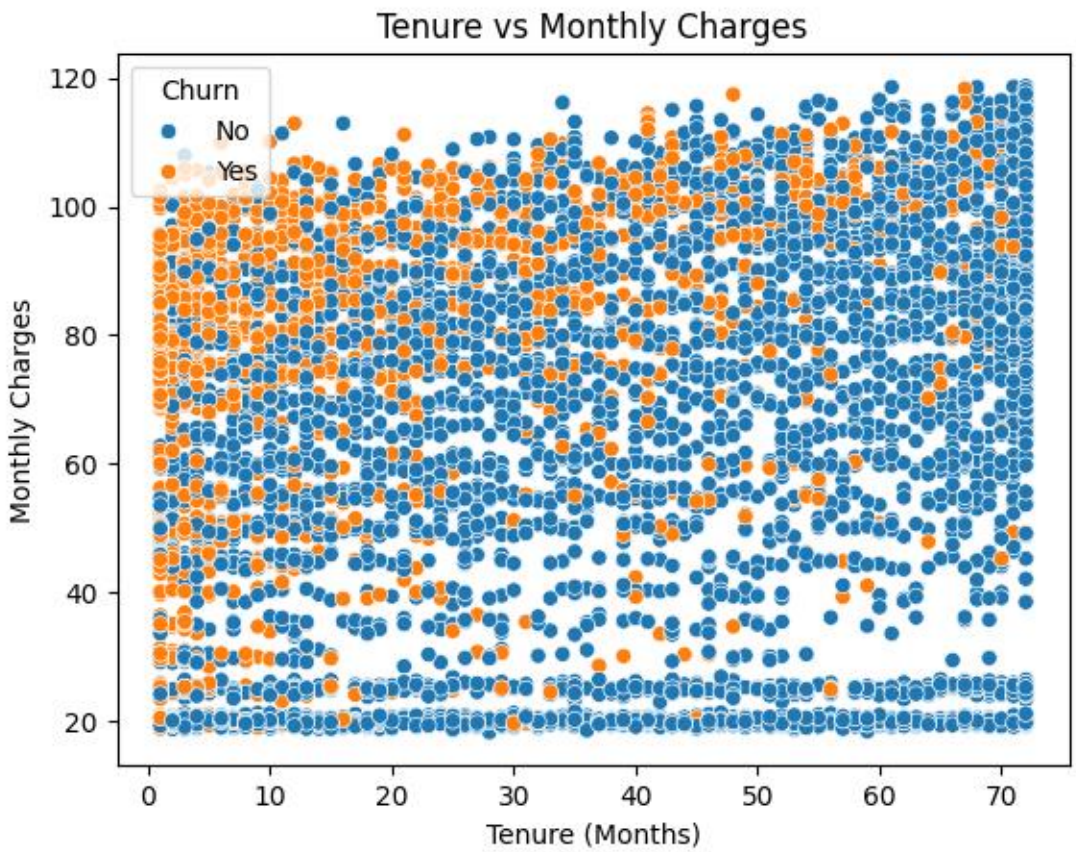
- Customers who churn have a higher median monthly charge.
- This indicates pricing plays a major role in churn.



GRAPH 5

6. Tenure vs Monthly Charges (Scatter Plot)

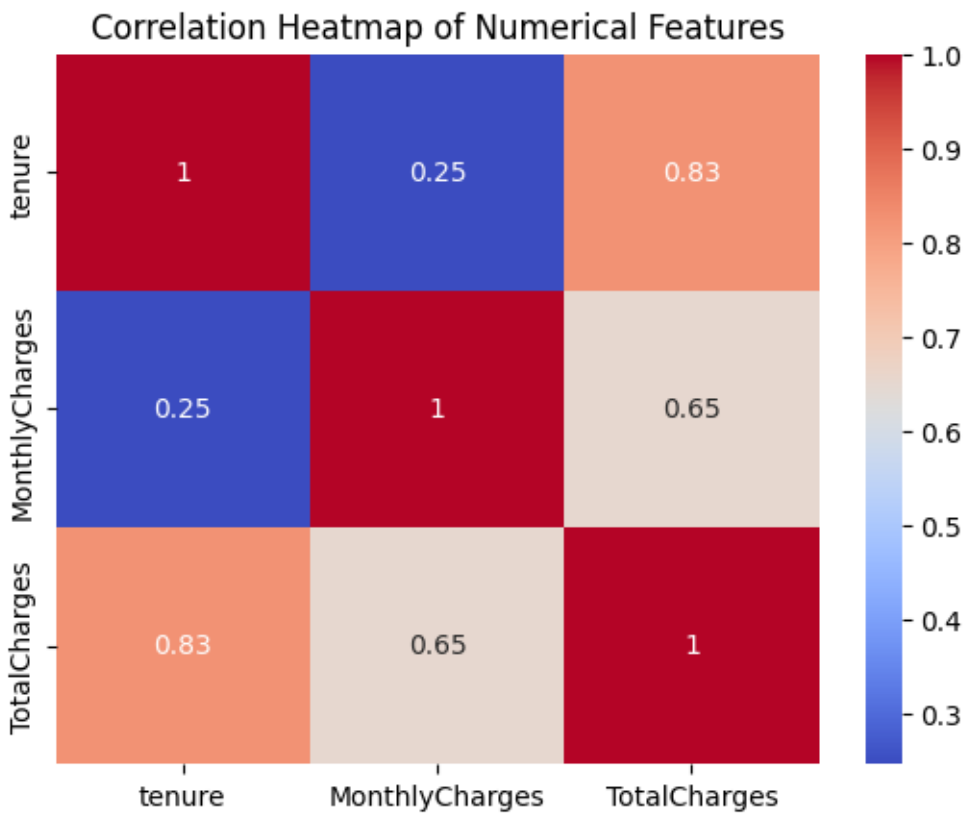
- Customers with low tenure and high charges are more likely to churn.
- Long-term customers show lower churn.



GRAPH 6

7. Correlation Heatmap

- Strong correlation between tenure and total charges.
- Weak correlation between tenure and monthly charges.



GRAPH 7

Key Findings

- A significant portion of customers (~26%) have churned, indicating a notable customer retention issue
- Customers with month-to-month contracts show the highest churn, while long-term contracts have much lower churn
- Electronic check users have the highest churn rate compared to other payment methods
- Customers with higher monthly charges are more likely to churn, as seen in the box plot
- Fiber optic internet users show higher churn compared to DSL and no internet users
- Short-tenure customers are more likely to churn, as observed in the scatter plot

- Gender has no significant impact on churn behaviour.

Patterns Discovered

- Customers with low tenure + high monthly charges form the highest churn group
- Long-term contracts (1–2 years) strongly reduce churn probability
- Customers using manual payment methods (electronic check) are more likely to leave
- There is a strong positive relationship between tenure and total charges (confirmed by heatmap and pair plot)
- Monthly charges do not strongly depend on tenure, indicating pricing is independent of duration
- Customers who stay longer tend to become more stable and less likely to churn
- Churn is not influenced by a single factor but by a combination of pricing, tenure, and contract type

"The analysis reveals that customer churn is primarily driven by **short-term commitments, higher pricing, and low customer tenure**, highlighting the need for targeted retention strategies"

6. Conclusion

This project focused on analysing customer churn in a telecom company using the Telco Customer Churn dataset. The primary objective was to understand customer behavior and identify the key factors that influence customers to discontinue their services. Through a structured approach involving data cleaning, preprocessing, and exploratory data analysis, meaningful insights were derived from the dataset. Various visualization techniques such as bar charts, histograms, scatter plots, boxplots, and heatmaps were used to clearly interpret patterns and relationships between different variables.

The analysis revealed that customer churn is influenced by multiple factors rather than a single variable. One of the most significant findings is that customers with **month-to-month contracts** are more likely to churn compared to those with long-term contracts. Similarly, customers with **shorter tenure** show a higher tendency to leave, indicating that new customers are more vulnerable. Another important observation is that customers with **higher monthly charges** are more likely to churn, suggesting that pricing plays a crucial role in customer retention. Additionally, payment methods such as **electronic check** are associated with higher churn rates, while automatic payment methods contribute to better retention.

The project also highlighted that certain factors,such as **gender**, do not significantly affect churn behavior, whereas service-related aspects like internet type and customer engagement levels have a noticeable impact. Multivariate analysis further confirmed that churn is driven by a combination of factors such as tenure, pricing, and service usage rather than any single parameter. The correlation analysis showed strong relationships between tenure and total charges, reinforcing expected business behavior.

Overall, the project successfully achieved its objective of identifying key drivers of customer churn and understanding customer behavior patterns. The insights obtained can help businesses design effective strategies such as promoting long-term contracts, improving customer onboarding, optimizing pricing, and encouraging secure payment methods. By focusing on high-risk customer segments, companies can reduce churn, improve customer satisfaction, and enhance long-term profitability.

7. Future Scope

- Develop **predictive models** using machine learning algorithms such as Logistic Regression and Random Forest to forecast customer churn
- Perform **feature engineering** to enhance the quality of input variables and improve model performance
- Apply **hyperparameter tuning** techniques to optimize model accuracy and reliability
- Incorporate **larger and more diverse datasets** to improve generalizability of results
- Integrate **real-time data processing** to monitor customer behavior continuously
- Use **customer segmentation** methods to identify and target high-risk customer groups
- Include **customer feedback and satisfaction data** to gain deeper behavioral insights
- Build **interactive dashboards** (using tools like Power BI or Tableau) for better business visualization
- Explore **advanced analytics techniques** such as clustering and time-series analysis
- Deploy the solution as a **decision-support system** to assist businesses in implementing effective retention strategies

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Appendix

Code:

```
# import libraries


import pandas as pd

import matplotlib.pyplot as plt

import numpy as np

import seaborn as sns


# load dataset

df = pd.read_csv("/content/WA_Fn-UseC_-Telco-Customer-Churn.csv")

df.head(10)          # shows first 10 rows

df.shape             #  Show Dataset Size (rows, columns)

df.columns           # Show Column Names

df.info()            # basic information about the dataset structure.

df.describe()        # statistical summary of numerical columns.

df.isnull().sum()    # To check null values


# Convert TotalCharges to numeric
```

```
df["TotalCharges"] = pd.to_numeric(df["TotalCharges"], errors="coerce")

df = df.dropna()        # Remove Missing Rows

df.shape

df.info()


# Churn Distribution

ax = sns.countplot(x="Churn", data=df)

plt.title("Customer Churn Distribution")

plt.xlabel("Churn")

plt.ylabel("Number of Customers")

for i in ax.containers:

    ax.bar_label(i)

plt.show()


# Contract Type Distribution

ax = sns.countplot(x="Contract", data=df)

plt.title("Distribution of Contract Types")

plt.xlabel("Contract Type")

plt.ylabel("Number of Customers")

for container in ax.containers:

    ax.bar_label(container)
```

```
plt.show()


# Contract Type Distribution
ax = sns.countplot(x="Contract", hue="Churn", data=df)
plt.title("Churn by Contract Type")
plt.xlabel("Contract Type")
plt.ylabel("Number of Customers")
for container in ax.containers:
    ax.bar_label(container)
plt.show()


# Churn by Gender
ax = sns.countplot(x="gender", hue="Churn", data=df)
plt.title("Churn by Gender")
plt.xlabel("Gender")
plt.ylabel("Number of Customers")
for container in ax.containers:
    ax.bar_label(container)
plt.show()


df["SeniorCitizen"] = df["SeniorCitizen"].replace({0: "Not Senior Citizen", 1: "Senior Citizen"})
```

```
# Churn by Senior Citizen
ax = sns.countplot(x="SeniorCitizen", hue="Churn", data=df)
plt.title("Churn by Senior Citizen")
plt.xlabel("Customer Type")
plt.ylabel("Number of Customers")
for container in ax.containers:
    ax.bar_label(container)
plt.show()

# Internet Service vs Churn
ax = sns.countplot(x="InternetService", hue="Churn", data=df)
plt.title("Churn by Internet Service Type")
plt.xlabel("Internet Service Type")
plt.ylabel("Number of Customers")
for container in ax.containers:
    ax.bar_label(container)
plt.show()

# Monthly Charges Distribution
sns.histplot(df["MonthlyCharges"], bins=30)
```

```
plt.title("Distribution of Monthly Charges")
plt.xlabel("Monthly Charges")
plt.ylabel("Number of Customers")
plt.show()

# Tenure Distribution
sns.histplot(df["tenure"], bins=30)
plt.title("Distribution of Customer Tenure")
plt.xlabel("Tenure (Months)")
plt.ylabel("Number of Customers")
plt.show()

# Payment Method Distribution
plt.figure(figsize=(8,5))
sns.countplot(x="PaymentMethod", hue="Churn", data=df)
plt.title("Churn by Payment Method")
plt.xlabel("Payment Method")
plt.ylabel("Number of Customers")
plt.xticks(rotation=45)
plt.show()
```

```
# Monthly Charges vs Churn
sns.boxplot(x="Churn", y="MonthlyCharges", data=df)
plt.title("Monthly Charges vs Churn")
plt.xlabel("Churn")
plt.ylabel("Monthly Charges")
plt.show()

# Tenure vs Monthly Charges
sns.scatterplot(x="tenure", y="MonthlyCharges", hue="Churn", data=df)
plt.title("Tenure vs Monthly Charges")
plt.xlabel("Tenure (Months)")
plt.ylabel("Monthly Charges")
plt.show()

# pair plot
sns.pairplot(df[["tenure","MonthlyCharges","TotalCharges"]])
plt.show()

# Correlation Heatmap
corr = df[["tenure", "MonthlyCharges", "TotalCharges"]].corr()
sns.heatmap(corr, annot=True, cmap="coolwarm")
```

```
plt.title("Correlation Heatmap of Numerical Features")

plt.show()
```